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Seyed Moeen Nehzati

Education

2022–2027 **PhD in Economics**.
New York University, NY, USA

2015–2019 **BS in Computer Engineering**.
Sharif University of Technology, Tehran, Iran

Research Interest

Optimization - Optimal Transport - Neural Networks - Experimental Design.

Work Experience

2020-2022 **Predoctoral research fellow**, *UCLA and USC*.
I worked on disentangling the complex relationship between genetic factors and behavioral traits. I was responsible for solving statistical models and implementing efficient solutions scalable to terabytes of data. This effort led to papers published in Nature and the development of statistical software regularly used by practitioners in the field.

2019 **Data Scientist**, *Cafe bazaar*.
Cafe Bazaar is an Iranian Android marketplace with more than 50 million active users. There, I worked on optimizing the advertising auctions and evaluating them empirically. Due to my technical background, I also participated in the creation, implementation, and deployment of the machine learning models needed for resolving auctions.

Research

Published

2026 **"Universal Representation of Generalized Convex Functions and their Gradients"**, *ICML*, *Moeen Nehzati*.

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2025 **"Family-based genome-wide association study designs for increased power and robustness"**, *Nat. Genet.* *Junming Guan et al.*.

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2022 **"Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals"**, *Nat. Genet.* *Aysu Okbay et al.*.

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2022 **"Mendelian imputation of parental genotypes improves estimates of direct genetic effects"**, *Nat. Genet.* *Alexander Young et al.*.

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Preprints

2026 **"Inference From Random Restarts"**, *arXiv*, *Moeen Nehzati and Diego Cussen*.

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2024 **"Family-GWAS reveals effects of environment and mating on genetic associations"**, *medrXiv*, *Tammy Tan et al.*.

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2021 **"Building Stable Off-chain Payment Networks"**, *arXiv*, *MohammadAmin Fazli, Moeen Nehzati, and MohammadAmin Salarkia*.

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Work in progress

2026 **"Nash-in-Nash Equilibrium with Inefficient Contracts"**, *Diego Cussen and Moeen Nehzati*.

2026 **"Compelled Conclusions Over Model Classes: Beyond Knife-Edge Degeneracies"**, *Moeen Nehzati and Diego Cussen*.

Software

famulus, *available here*.

A cross-LLM personal and research assistant plugin for Claude Code and Codex, combining model-interpreted skills with machine-executable Python so routine steps don't depend on model judgment. It serves as a personal and research assistant, handling tasks like planning the day, reviewing emails, and auditing proofs and bibliographies, among others.

gconvex, *available here*.

Implements Finitely Convex Models developed in Universal Representation of Generalized Convex Functions and their Gradients (https://arxiv.org/abs/2509.04477). The models are used to find optimal selling mechanism with single buyer and multiple goods. The results match the theoretical optimal benchmark and are quick to be found even on personal computers.

snipar, *available here*.

snipar (single nucleotide imputation of parents) is a Python package for inferring identity-by-descent (IBD) segments shared between siblings, imputing missing parental genotypes, and for performing family based genome-wide association and polygenic score analyses using observed and/or imputed parental genotypes. The imputation method and the family-based GWAS and polygenic score models are described in Young et al. 2022.

poissonreg, *available here*.

A simple library for sparse poisson regression, based on PyTorch.

Technical skills

Python/Java/Cython Expert

Scientific Programming Expert

Pytorch/Scikit/Spark Experienced

C/C++/R/SQL/Julia Experienced

Awards and Honors

Among top 0.05% in Iran's national university entrance exam.

Member of Iran's National Elites Foundation.

Languages

Persian(native)/ English(Fluent)/ Arabic(basic)

References

name	email address
Prof. Alfred Galichon Courant Institute of Mathematical Sciences & Department of Economics, NYU	ag133@nyu.edu
Prof. David Cesarini Department of Economics, NYU	dac12@nyu.edu
Prof. Daniel Benjamin Anderson School of Management, UCLA	daniel.benjamin@anderson.ucla.edu